

# Prediction-based limit order trading

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**Abstract**—Managing the trade-off between volume and margin is among the most fundamental challenges for dealers in a securities market. We attempt to overcome this trade-off by incorporating predictions for buyer- and seller-initiated trades when submitting limit orders. Using the Avellaneda–Stoikov model as an example, we show how dealers can adjust quotes to predictions and thereby capture larger spreads at constant volume. Simulations on historical limit order book data illustrate that our model allows dealers to both increase market making revenues through trade flow-optimized positioning in the order book and reduce adverse selection cost through preempted adverse price movements.

**Keywords**—Sequence learning, Limit order trading, Avellaneda–Stoikov model, Adverse price selection, Cryptocurrency.

**JEL classification**—C45, C53, C55, C63, D49, G17.

## 1 Introduction

Intuitively, dealers aim to repeatedly earn the ‘spread’ between buy and sell prices by simultaneously buying and selling an asset without accumulating a large net position. As a result, dealers face a trade-off between trade volume and margin: Wide spreads lead to high margins but less volume, while dealers with tight spreads trade frequently but at unfavorable prices [31]. Additionally [34], dealers face inventory risk arising from the stochastic behavior of the net position [5], execution risk arising from limit orders not being executed or only partially executed [36, 40], and adverse selection risk arising from trading with informed traders [23]. Thus, much of the literature has been concerned with the question of whether there is an optimal strategy for posting limit orders [2].

Avellaneda and Stoikov [5] propose a stochastic optimal control model in which the dealer’s reservation price is a function of the net position: When the dealer is long (short), the reservation price is below (above) the mid-price to incentivize execution of sell (buy) orders and return to a neutral inventory, thereby allowing the dealer to manage inventory risk. Numerous studies [2, 6, 7, 9, 10, 12, 22, 31, 32, 34] discuss extensions of the model. Like [5], most of these studies assume that market orders follow a stochastic process. In contrast, [8, 17, 18, 39, 62, 63, 66, 67] present evidence of mid-price predictability from the limit order book (LOB). Moreover, [37] finds that high-frequency traders anticipate specific order flow trends, while [22] shows that high-frequency traders can improve their P&L with directional bets.

Motivated by these discussions, we introduce an approach to integrate trade flow forecasting into limit order trading. We model buyer- and seller-initiated trades using a Recurrent Neural Network (RNN) – a powerful sequence learner well-suited for modeling the high-frequency dynamics of an LOB [19] – that processes LOB and trade data to predict average 24-hour percentile ranks of buyer- and seller-initiated trade sizes during the next five seconds. For example, a 90% prediction for buyer-initiated trades means that the average buy trade size over the next five seconds is greater than 90% of all buyer-initiated trades over the last 24 hours.

Using the Avellaneda–Stoikov model as an example, we adjust limit order prices to these predictions in two ways: First, if our model predicts large (small) trades, we increase (decrease) the depth of orders in the LOB, i.e., post orders deeper (closer) in the

LOB. For example, when predicting large buy orders and small sell orders, we increase both the ask and bid price to maximize the potentially captured spread on the ask side and increase the chance of being consumed on the bid side. Second, we shift the reservation price to protect against the anticipated price movement, which we derive from the imbalance between the predictions for buy and sell trade sizes. Hence, when anticipating a price increase (decline), we shift the reservation price up (down) to protect from being adversely selected by informed buyers (sellers) and participate in the price increase (decline) through long (short) exposure.

We test our model on historical LOBs of the BTC-USD and ETH-USD pairs and arrive at the following findings: The RNN improves prediction quality by about 40% over a standard feedforward neural network and performs best when previous trade flow is balanced. Moreover, there is an empirical optimum of sensitivity to predictions, which states that we should be twice as sensitive to predictions when adjusting order depth to trade sizes than when adjusting reservation price to adverse price movements. Our limit order model significantly improves P&L over a wide range of test periods compared to the Avellaneda–Stoikov model, e.g., by up to 7% for one-hour periods, but this improvement also comes with higher risk and depends on market dynamics. As two main contributors to the P&L, we identify higher market making revenues due to larger spreads earned especially when the order book is less liquid, and reduced adverse selection costs by preempting adverse price movements, with the cost reduction being proportional to the magnitude of the adverse price movement. Accordingly, our model provides an effective approach to both overcoming the trade-off between volume and margin and reducing adverse selection cost by taking trade flow forecasts into account when deciding whether we can afford a wider spread to increase profitability.

## 2 High-frequency trading in a limit order book

A limit order is an order to buy or sell a specified amount of an asset at a specified price. If execution is not possible at the time of arrival, the order queues in the LOB at the respective price level. The LOB typically follows a price-time priority, with orders queued according to their price level and arrival time. For more information on LOBs, we refer to [1, 29]. A market order, in contrast, has no price limit and serves impatient traders by instantaneous execution against open orders in the LOB, starting with the best price level and, if necessary, continuing with subsequent price levels until the total order volume has been matched. The objective of dealers is to provide liquidity to the market by continuously offering buy and sell prices through the LOB without accumulating large net positions [47]. The prices offered therefore should take into account a dealer’s net position: Symmetrically placing orders, e.g., at the top of the order book, risks adverse selection, as downward or upward trends could lead to a disproportionate number of filled buy or sell orders, respectively, resulting in a large net position that the dealer would have to offset at unfavorable prices.

A common approach in the literature for determining a dealer’s optimal buy and sell price is to treat market making as a stochastic optimal control problem introduced by [38], who analyzes the optimal price of a single stock for a monopolistic dealer. Avellaneda and Stoikov [5] extend this approach and formulate the

problem mathematically for a dealer who optimizes the expected utility of the terminal profit. Consistent with empirical indications [24, 28, 46, 50, 64], the authors assume that market order flow follows a Poisson process with rate  $Ae^{-\kappa\delta}$ , where  $\delta$  is the distance between order price and mid-price, and  $A$  and  $\kappa$  represent order book liquidity. The authors derive the indifference or reservation price  $r_{AS}$  and the optimal bid-ask spread  $\delta^a + \delta^b$  as

$$r_{AS} = s - q\gamma\sigma^2(T - t) \quad (1)$$

$$\delta^a + \delta^b = \gamma\sigma^2(T - t) + \frac{2}{\gamma}\ln\left(1 + \frac{\gamma}{\kappa}\right) \quad (2)$$

where  $s$  denotes the mid-price,  $q$  depicts the inventory net position,  $\sigma$  is the standard deviation of the mid-price,  $\gamma$  is an inventory risk aversion parameter, and ‘AS’ indicates the Avellaneda–Stoikov model. The optimal ask price is  $r_{AS}^a = r_{AS} + (\delta^a + \delta^b)/2$  and the optimal bid price is  $r_{AS}^b = r_{AS} - (\delta^a + \delta^b)/2$ . Close to terminal,  $\delta^a + \delta^b$  converges towards  $2/\gamma \ln(1 + \gamma/\kappa)$  and becomes symmetrical to the mid-price, increasing the dealer’s chance of finishing with a zero net position. The ask and bid order’s depth in the order book is  $\delta_{AS}^a = r_{AS}^a - s$  and  $\delta_{AS}^b = s - r_{AS}^b$ , respectively.

A large literature discusses extensions of the Avellaneda–Stoikov model [2, 6, 7, 9, 10, 12, 22, 31, 32, 34]. Studies focusing on re-modeling the market order flow include [2, 22], both of which, inline with [13], replace the Brownian process with a mean-reverting process, and [7], who considers the liquidation problem and models order arrivals as a function of the liquidation price. More recently, reinforcement learning has led to improvements over stochastic optimal control formulations – see [25, 26, 33, 43, 56, 60]. On the question of whether market order flow is stochastic, numerous studies examine the relationship between LOBs and price discovery. Yet, [18] acknowledges that there is “no consensus on the extent to which LOBs convey predictive information”. While [27] and [58] argue that limit orders are not informative, [8, 17, 18, 39, 62, 63, 66, 67] state that LOBs contain predictive information. For more studies, we refer to the survey of [65]. [42] uses sentiment predictions from LOBs and market news for limit order trading, whereas [35] implements price predictions into a reinforcement learning-based limit order model.

### 3 Data

[56] outlines that cryptocurrency exchanges “represent a well-suited test environment” for limit order trading models, as – in contrast to global stock exchanges – data are easily and free to access and “major cryptocurrency exchanges operate like established limit order exchanges and exhibit very similar stylized facts” [45, 57]. As one of the most liquid centralized cryptocurrency exchanges with more than 100 million traders [16], Coinbase is therefore a suitable example to test our limit order model. Accordingly, we focus on the two most liquid pairs, ‘BTC-USD’ and ‘ETH-USD’.

Since tick-level data from Coinbase is not downloadable, we record LOB and trade messages between 2022-08-16 05:37:27 UTC and 2022-10-05 16:07:40 UTC via WebSocket using cbpro [49], the Python client for the Coinbase Pro API. We receive batches of level-2 messages every 50 milliseconds through the level2\_batch channel and price updates on each match through the ticker channel. After recording, we replay the messages to reconstruct LOBs and trade flows, aggregate both data sets into one-second intervals and cut the LOBs behind the top 50 price levels on each side of the book. To obtain comparable results between the two pairs, we index all prices at 100 with the first recorded price.

For the split into training, validation and test periods, we follow the reasoning of [21] and use  $n$ -fold cross validation that divides the entire period into  $n$  equal-sized sub-intervals, for each

of which we train independent models. We choose an overlapping split of five days in length, of which the first three days are for training (60%), the fourth day is for validation (20%), and the last day is for test (20%). The overlap is four days, i.e., we shift the sub-intervals by one day, yielding a total of 45 sub-intervals, allowing us to test each day once with an independent model.

## 4 High-frequency trade flow prediction

### 4.1 24-hour trade size percentile ranks

We model buyer- and seller-initiated trade flows using the percentile ranks of trade sizes in the frequency distribution of all trade sizes over the past 24 hours. We calculate the 24-hour percentile rank  $R$  of a trade size as the share of trade sizes that are less than that trade size, indicating whether the trade tends to be large or small relative to other trades from the previous 24 hours. Thus,  $R$  is defined in the range  $[0, 1]$ , with  $R > 0.5$  ( $R < 0.5$ ) indicating a relatively large (small) trade. Considering the differences between bid and ask market order flow, we calculate percentile ranks separately for buyer-initiated trades ( $R^b$ ) and seller-initiated trades ( $R^a$ ). For example, if a buyer-initiated trade about 1.5 BTC has a percentile rank of  $R^b = 0.90$ , this means that 90% of all buyer-initiated trades over the past 24 hours are smaller than 1.5 BTC. To obtain comparable results across all sub-intervals, we calculate percentile ranks prior to separating training, validation and test periods. Figure 5 illustrates that, although we calculate percentiles based on sequences along the time axis, the percentile ranks are evenly distributed across test periods.

Together,  $R^b$  and  $R^a$  convey two pieces of information: First,  $R^b$  and  $R^a$  indicate whether trades are more likely to consume one or multiple price levels. Figure 14 in the Appendix illustrates that trades with a low (high) trade size percentile rank consume one (multiple) price level(s), e.g., trade sizes with  $R = 90\%$  consume five price levels on average. Second, the difference between  $R^b$  and  $R^a$  indicates the imbalance between the buyer- and seller-initiated trade flow. For example, a small  $R^b$  combined with a large  $R^a$  may represent a sell surplus. In contrast to the widely used definition [3, 59], we follow [54] and define the trade flow imbalance  $TFI$  in the range  $[-1, 1]$  by  $(R^b - R^a)/(R^b + R^a)$ , where  $TFI < 0$  ( $TFI > 0$ ) indicates a sell (buy) surplus. Consistent with [15, 59], we consider  $TFI$  as an approximation of the price movement, with  $TFI > 0$  ( $TFI < 0$ ) indicating a price increase (decline).

### 4.2 Sequence learning for limit order trading

We present a limit order trading model that adjusts buy and sell prices according to the predicted 24-hour percentile rank of future seller- and buyer-initiated trade sizes. The prediction task is a regression task with  $R^b$  and  $R^a$  as target variables. As [59] shows that trade flow imbalance is auto-correlated and [18] states that high-frequency price “increments are neither independent nor stationary and depend on the state of the order book”, we assume that the observations of  $R^b$  and  $R^a$ , respectively, are auto-correlated and not i.i.d. Hence, we define our prediction task as a sequence prediction model that uses lagged values of  $R^b$  and  $R^a$  sequences to predict subsequent values of that sequences. A sequence learner proven successful in modeling high-frequency dynamics is the RNN [19]. While some researchers [63, 66] prefer Long Short-term Memories (LSTM) over RNNs, [19] outlines that for high-frequency LOB modeling, “short-term memory is adequate”, i.e., the auto-correlation lowers for larger lags and the vanishing gradient problem of RNNs [53] is not an issue, and hypothesizes that “there is little to no benefit in using an LSTM”.

Introduced by [52], the RNN is an artificial neural network architecture capable to extract temporally encoded information from time-series data by using feedback connections between neurons. A hidden state transfers information contained in cell out-

puts across time steps and facilitates connection of information between time steps, thereby allowing past information to persist. The output of an RNN cell at time  $t$  is the vector of hidden states  $h_t$ . At each time step  $t$ , the cell processes both  $h_{t-1}$  and the vector of current elements of the input sequence,  $x_t$ , to generate the new hidden state  $h_t = \tanh(w_x x_t + w_h h_{t-1} + b_h)$  through a  $\tanh$  activation function, where  $w_x$  ( $w_h$ ) is a vector of weights determining which elements of  $x_t$  ( $h_{t-1}$ ) to keep and which to remove. This procedure ensures that the network considers important knowledge acquired in previous time steps when processing new information. The model output at time  $t$  is  $y_t = w_y h_t + b_y$ . In our case,  $y_t$  is a vector containing  $\hat{R}_t^b$  and  $\hat{R}_t^a$ .

### 4.3 Predictors of trade flow

Besides the sequences of  $R_t^b$  and  $R_t^a$ , we additionally include sequences of predictors proposed by the price prediction literature: First, we include the price level distance, which is the distance of each of the 10 best price levels on both sides of the order book from the mid-price, expressed as a percentage of the mid-price [63]. Second, we include the cumulative depth size, which reports the cumulative sum of open limit order sizes at each of the 10 best price levels [63]. Both variables yield a vector of 20 unique values for each time step. Third, we include the order flow imbalance *OFI* [17, 48], which we calculate as the difference between the cumulative bid and ask size at the tenth price levels on each side, divided by their sum [11]. For example, if the cumulative size at the 10th price level on the bid side is 5 ETH and on the ask side 4 ETH, it is  $OFI = (5 - 4)/(5 + 4) = 0.11$ . Finally, we include the percentage change in the mid-price [63, 65] and the distance between the highest and lowest trade price of all matches within each one-second interval [65].

In total, given that we process  $n$  observations per batch, each time step includes five predictors with shape  $n \times 1$  and two predictors with shape  $n \times 20$ , i.e., we process 45 predictors per time step. Motivated by [18] who chooses a sequence length of 10 for tick-level data, we use a sequence length of 10 seconds, i.e., we provide ten lags to our model in order to deliver a prediction. The final vector of input variables thus has shape  $n \times 45 \times 10$ . Prior to training, we standardize the input data by subtracting the mean and dividing by the standard deviation across all sub-intervals [41].

### 4.4 Network configuration

We use a three-layer network architecture with two visible layers and one hidden layer. The number of units in the first and last layer equals the dimension of the vector of input and output variables, respectively. The activation function for the two output units is the sigmoid function, which squashes predictions into the range  $[0, 1]$ , corresponding to the range for which  $R^b$  and  $R^a$  are defined. Determining the number of units in the hidden layer is not as straightforward. Out of architectures with 16, 32, 64, 128, and 256 hidden units, we choose 64 hidden units, resulting in the lowest mean squared prediction error for the validation sets. As activation function we select  $\tanh$ , scaling values between  $[-1, 1]$  and thereby avoiding dropping information by scaling towards 0.

Each training step, we minimize the mean squared prediction error, a loss function suited for regression problems in modeling high-frequency dynamics [20], across batches of 8 random samples using the stochastic gradient descent [30, 51] with an initial learning rate of 0.01. We regularize training in multiple ways: First, we consider a stateless RNN, i.e., memory does not transfer between sequences of different batches and persist for one batch. Second, we shuffle samples within batches. Third, we apply 20% dropout [61] to the input layer and, as proposed by [30], the non-recurrent connections in the hidden layer. Finally, we use early stopping and halt training when there has been no improvement for the validation set for two epochs.

## 5 Prediction-based limit order trading

### 5.1 Prediction-based order depth

We propose order depth as a function of the predicted trade size percentile ranks,  $\hat{R}^b$  and  $\hat{R}^a$ . Assumed that we make predictions one period ahead and use these predictions to adjust  $r_{AS}^a$  and  $r_{AS}^b$ , respectively, the function should fulfill two requirements: First, if  $\hat{R}^b < 0.5$  ( $\hat{R}^a < 0.5$ ), i.e., predicting small buy (sell) orders, we should post sell (buy) orders closer to the top of the book to increase the hit probability, whereas if  $\hat{R}^b > 0.5$  ( $\hat{R}^a > 0.5$ ), we should post sell (buy) orders deeper in the book to maximize the captured spread. Second, the spread adjustment should be proportional to the percentile rank, i.e., the adjustment for  $\hat{R}^b = 0.9$  should be larger than that for  $\hat{R}^b = 0.6$ . Based on the Avellaneda–Stoikov model, we define the spread-adjusted ask price  $r_{SP}^a$  and the spread-adjusted bid price  $r_{SP}^b$  by

$$r_{SP}^a = r_{AS} + \frac{1}{2} \left( 1 + \frac{1}{\alpha_1} (2\hat{R}^b - 1) \right) (\delta^a + \delta^b) \quad (3)$$

$$r_{SP}^b = r_{AS} - \frac{1}{2} \left( 1 + \frac{1}{\alpha_1} (2\hat{R}^a - 1) \right) (\delta^a + \delta^b) \quad (4)$$

where  $\alpha_1 \geq 0$  is a scaling parameter that determines how sensitive order depth is to predicted percentile ranks, i.e., by how much we widen or tighten the spread in response to a prediction. As equation (3) can be rewritten as  $r_{SP}^a = r_{AS}^a + (\delta^a + \delta^b)(2\hat{R}^b - 1)/(2\alpha_1)$ , we simply add a weighted version of  $\delta^a + \delta^b$  to  $r_{AS}^a$ . The terms  $(2\hat{R}^b - 1)/\alpha_1$  and  $(2\hat{R}^a - 1)/\alpha_1$  are positive for  $\hat{R}^b > 0.5$  and  $\hat{R}^a > 0.5$  as we wish to post orders deeper in the LOB, zero for  $\hat{R}^b = 0.5$  and  $\hat{R}^a = 0.5$  as predicted orders are neither large nor small and we are indifferent between widening or tightening spreads, and negative for  $\hat{R}^b < 0.5$  and  $\hat{R}^a < 0.5$  as we wish to narrow the spread. For  $\alpha_1 = 1$ , the terms are defined in the range  $[-1, 1]$ , implying that the maximum spread adjustment ( $\hat{R}^b, \hat{R}^a = 1$ ) is  $r_{SP}^a - r_{AS}^a = (\delta^a + \delta^b)/2$  and  $r_{AS}^b - r_{SP}^b = (\delta^a + \delta^b)/2$ , i.e., the order's depth in the order book is twice the depth given by the Avellaneda–Stoikov model, and the minimum spread adjustment ( $\hat{R}^b, \hat{R}^a = 0$ ) is  $r_{SP}^a - r_{AS}^a = -(\delta^a + \delta^b)/2$  and  $r_{AS}^b - r_{SP}^b = -(\delta^a + \delta^b)/2$ , i.e.,  $r_{SP}^a, r_{SP}^b = r_{AS}$ . Smaller (larger) values for  $\alpha_1$  are associated with more (less) impact of predictions on the depth.

Figure 1 illustrates  $r_{SP}^a - r_{AS}^a$  (left) and  $r_{SP}^b - r_{AS}^b$  (right) as functions of  $\alpha_1$  and  $\hat{R}^b$  and  $\hat{R}^a$ , respectively. For illustration purposes, we set  $s = 100$ ,  $\gamma = 0.25$ ,  $\sigma = 0.3$ ,  $T = 1$ ,  $t = 0.3$ ,  $\kappa = 2$ ,  $q = 0$ . Both charts emphasize that small (large) values for  $\alpha_1$  are associated with strong (little) changes to the quotes as order depth reacts more (less) strongly to predictions. If  $\hat{R}^b > 0.5$ , we increase the ask price to exploit a larger spread from arriving large buy orders, and lower the ask price if  $\hat{R}^b < 0.5$  to increase the chance of being consumed by small buy orders. Correspondingly, we decrease the bid price if  $\hat{R}^a > 0.5$  and increase it if  $\hat{R}^a < 0.5$ . One may interpret our extension as a way to reduce adverse selection cost: Informed trades typically are larger than uninformed trades [14, 44]. By increasing spreads when predicting large orders, we potentially protect against adverse selection by informed traders [55].

### 5.2 Prediction-based reservation price

Adverse price selection occurs when better informed traders pick off our limit orders on one side of the order book [9], resulting in an adverse price movement for our limit orders on the other side, which now queue away from the inside market. Following [18], one approach “potentially reducing the likelihood of adverse price movement” between the fill of a sell and a buy order is predicting price movements, thereby preempting unfavorable execution of orders. For example, [9] simulates an informed dealer using a hidden Markov model and reports reduced adverse selection costs.

We propose the reservation price as a function of the predicted trade flow imbalance,  $\widehat{TFI}$ , given by  $\widehat{TFI} = (\hat{R}^b - \hat{R}^a)/(\hat{R}^b + \hat{R}^a)$ .

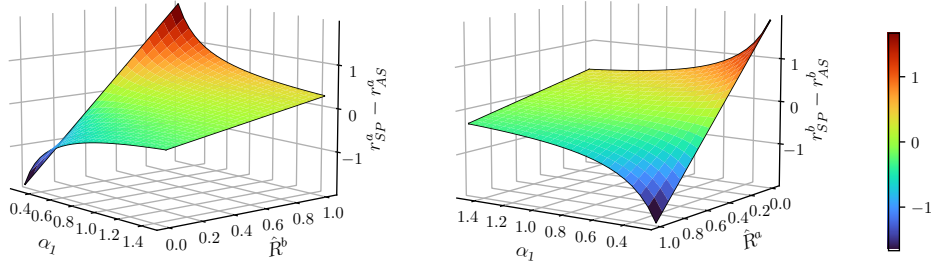


Figure 1: Prediction-based spread adjustments

This function should fulfill two requirements: First, if  $\widehat{TFI} > 0$ , we should increase the reservation price in order to avoid adverse filling of sell orders and incentivize long exposure for participating in the price increase, whereas if  $\widehat{TFI} < 0$ , we should decrease the reservation price for avoiding adverse filling of buy orders and incentivize short exposure. Second, considering our net position, these directional adjustments should be allowed to contradict with the Avellaneda–Stoikov model, i.e., if we predict a price increase (decline), we should be able to use a reservation price greater (less) than  $r_{AS}$  even if we are long (short). We define the directional forecast-adjusted reservation price  $r_{DF}$  by

$$r_{DF} = s + \left( \text{sgn}(q) \frac{1}{\alpha_2} \widehat{TFI} - 1 \right) q \gamma \sigma^2 (T - t) \quad (5)$$

where  $\text{sgn}(q)$  is the signum function of  $q$  and  $\alpha_2 \geq 0$  measures how sensitive the reservation price is to a prediction. Rewriting equation (5) as  $r_{DF} = r_{AS} + \text{sgn}(q)(\widehat{TFI}/\alpha_2)q\gamma\sigma^2(T-t)$  illustrates that we simply add or subtract a weighted version of the Avellaneda–Stoikov correction term  $q\gamma\sigma^2(T-t)$ . For  $\alpha_2 = 1$  and  $q > 0$ , the maximum adjustment ( $\widehat{TFI} = 1$ ) is  $r_{DF} - s = 0$ , i.e., we neutralize the Avellaneda–Stoikov correction, and the minimum adjustment ( $\widehat{TFI} = -1$ ) is  $r_{DF} - s = 2(r_{AS} - s)$ , i.e., we double the distance to the mid-price. Smaller (larger) values for  $\alpha_2$  imply a higher (lower) price sensitivity to  $\widehat{TFI}$ . For instance, if  $\widehat{TFI} = 0.25$ , the price adjustment is  $2q\gamma\sigma^2(T-t)$  for  $\alpha_2 = 0.5$  and  $0.5q\gamma\sigma^2(T-t)$  for  $\alpha_2 = 2$ .

Figure 2 illustrates the difference between  $r_{DF}$  and  $r_{AS}$  for values in the range  $q = [-2, 2]$  and  $\widehat{TFI} = [-1, 1]$  with  $\alpha_2 = 1$ ,  $s = 100$ ,  $\gamma = 0.25$ ,  $\sigma = 0.3$ ,  $T = 1$ ,  $t = 0.3$ ,  $\kappa = 2$ . If predicting a sell surplus, it is  $r_{DF} < r_{AS}$ , either contradicting ( $q < 0$ ) or reinforcing ( $q > 0$ ) the Avellaneda–Stoikov correction. Analogous, if predicting a buy surplus, it is  $r_{DF} > r_{AS}$ , either contradicting ( $q > 0$ ) or reinforcing ( $q < 0$ ) the Avellaneda–Stoikov correction. The directional forecast-adjusted ask and bid quote is  $r_{DF}^a = r_{DF} + (\delta^a + \delta^b)/2$  and  $r_{DF}^b = r_{DF} - (\delta^a + \delta^b)/2$ , respectively.

### 5.3 Consolidated prediction-based limit order model

Finally, we combine both  $DF$  and  $SP$  models to one consolidated limit order model that considers  $r_{DF}$  as reservation price and determines bid and ask quotes based on equations (3) and (4) with

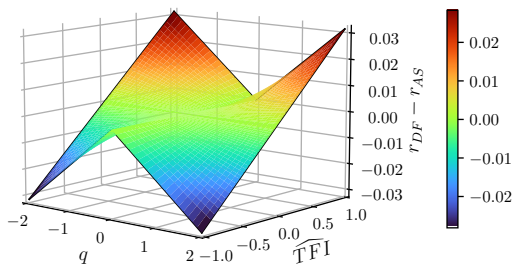


Figure 2: Prediction-based reservation price adjustments

replacing  $r_{AS}$  by  $r_{DF}$ . As a result, the consolidated ask price is  $r_{SPDF}^a = r_{DF} + (1 + (2\hat{R}^b - 1)/\alpha_1)(\delta^a + \delta^b)/2$  and the consolidated bid price is  $r_{SPDF}^b = r_{DF} - (1 + (2\hat{R}^a - 1)/\alpha_1)(\delta^a + \delta^b)/2$ .

One might wonder how  $DF$  and  $SP$  relate to each other and whether  $DF$  and  $SP$  adjustments to  $AS$  quotes would offset each other in the consolidated model. To understand their relationship, we compare the absolute difference between  $SPDF$  quotes and  $AS$  quotes (e.g.,  $|r_{SPDF}^a - r_{AS}^a|$ ) with the absolute difference between  $SP$  or  $DF$  quotes and  $AS$  quotes (e.g.,  $|r_{DF}^a - r_{AS}^a|$  or  $|r_{SP}^a - r_{AS}^a|$ ). By using absolute differences, we take into account that  $SP$  or  $DF$  corrections occur in both directions, either increasing or decreasing  $AS$  quotes, which allows us to summarize the comparison in four cases represented by inequations (6)–(9). For instance, if inequation (6) is true, this means that  $SPDF$  does not offset or reinforces the  $DF$  adjustment, whereas if it is false, this means that the  $SPDF$  adjustment to  $AS$  is less than the  $DF$  adjustment, i.e.,  $SPDF$  contradicts  $DF$  and neutralizes (parts of) the  $DF$  adjustment.

$$|r_{SPDF}^a - r_{AS}^a| \geq |r_{DF}^a - r_{AS}^a| \quad (6)$$

$$|r_{SPDF}^b - r_{AS}^b| \geq |r_{DF}^b - r_{AS}^b| \quad (7)$$

$$|r_{SPDF}^a - r_{AS}^a| \geq |r_{SP}^a - r_{AS}^a| \quad (8)$$

$$|r_{SPDF}^b - r_{AS}^b| \geq |r_{SP}^b - r_{AS}^b| \quad (9)$$

With regard to  $DF$  strategy, we find that inequation (6) is true if  $\hat{R}^b \geq \hat{R}^a$  and  $\hat{R}^b \geq 0.5$  or if  $\hat{R}^b \leq \hat{R}^a$  and  $\hat{R}^b \leq 0.5$ , and inequation (7) is true if  $\hat{R}^b \geq \hat{R}^a$  and  $\hat{R}^a \leq 0.5$  or if  $\hat{R}^b \leq \hat{R}^a$  and  $\hat{R}^a \geq 0.5$ . For instance, when  $DF$  shifts the reservation price upward ( $\hat{R}^b \geq \hat{R}^a$ ),  $SP$  does so as well if buy orders are relatively large ( $\hat{R}^b \geq 0.5$ ) or sell orders are relatively small ( $\hat{R}^a \leq 0.5$ ). Thus, if predicting a buy surplus, we increase the reservation price to protect from being adversely selected by informed buyers, thereby implicitly increasing (decreasing) the depth of buy (sell) orders in the LOB. This does not hold if the predicted buy surplus is based on a relatively small buy volume, as in that case, we choose a closer spread to increase the chance of being consumed by small orders.

With regard to  $SP$  strategy, we find that, analogous to  $DF$  strategy, inequation (8) is true if  $\hat{R}^b \geq \hat{R}^a$  and  $\hat{R}^b \geq 0.5$  or if  $\hat{R}^b \leq \hat{R}^a$  and  $\hat{R}^b \leq 0.5$ , and, inequation (9) is true if  $\hat{R}^b \geq \hat{R}^a$  and  $\hat{R}^a < 0.5$  or if  $\hat{R}^b \leq \hat{R}^a$  and  $\hat{R}^a \geq 0.5$ . If predicting large sell orders ( $\hat{R}^a \geq 0.5$ ) and a sell surplus ( $\hat{R}^b \leq \hat{R}^a$ ), we post buy orders deeper in the book to maximize the captured spread, thereby implicitly lowering the mid-point between ask and price, which is consistent with lowering the reservation in response to the sell surplus. If predicting a buy surplus instead, the  $DF$  adjustment would contradict the  $SP$  adjustment in response to  $\hat{R}^a \geq 0.5$ .

### 5.4 Trading model

We propose a limit order trading model in which we predict the five-second averages of  $R^a$  and  $R^b$ , use these predictions to determine bid and ask price, and then simultaneously place a corresponding buy order and a corresponding sell order. After five

seconds, the orders have either been filled by incoming market orders or we cancel them and submit new orders. Even if both orders get filled in less than five seconds, we still wait five seconds before submitting new orders. As target variables we consider the equal-weighted averages of all five values in the five-second intervals of  $R^b$  and  $R^a$ , respectively, while keeping the predictors in one-second granularity.

We follow [31] and base our trading model on several assumptions: First, the money market pays no interest and we can cancel and submit orders for free [5]. Second, there is no latency [34], which is a rather minor condition since we consider five-second intervals. Finally, our limit orders have no effect on the market [5, 12, 22, 60]. This is a realistic assumption as we model highly liquid currency pairs [66] and order sizes that are small compared to the total amount traded in the market [60]. Besides, we assume constant risk aversion  $\gamma = 0.5$  and order book density  $\kappa = 2$ , and compute  $\sigma$  over 24-hour windows, corresponding to  $R^b$  and  $R^a$ .

The size per order is 0.1 BTC and 0.1 ETH and the maximum net position is  $\pm 0.5$  BTC and  $\pm 0.5$  ETH, respectively. Once reached the inventory limit, we stop posting on the respective side until the net position is less than 0.5. We trade BTC and ETH simultaneously, with hedging of positions between BTC and ETH being not allowed, and calculate P&L following [5] separately for each currency pair. In the following sections, we report P&L as equally weighted average of both currency pairs. Since we use indexed prices, P&L does not represent currency units.

Due to the LOB's aggregate form, we have no access the time queue at a particular price level. While we can separate new from waiting orders based on trade history, if the size at a price level decreases without a trade at that time, we cannot identify which orders were cancelled. We follow [60] and assume that the probability that the canceled order is ahead our order is proportional to the volume ahead compared to the volume behind our order. Any increase (resp. decrease) in volume at a particular price level that is not related to a trade will queue behind our order (resp. be evenly distributed among the orders before and after our order).

### 5.5 Empirical calibration of $\alpha_1$ and $\alpha_2$

As we have historical data, we can empirically determine which values of  $\alpha_1$  and  $\alpha_2$  would have been P&L-optimal. For this calibration, we assume to have full knowledge of the future and determine quotes according to the realized  $R^b$  and  $R^a$ . Figure 3 illustrates the average one-hour P&L (solid lines with markers), i.e., the average P&L at the end of each hour, and the execution rate ExR (dashed lines), i.e., the share of executions in all submissions, achieved by *SP* as a function of  $\alpha_1$  (red) and achieved by *DF* as a function of  $\alpha_2$  (green). The blue lines represent the *AS* benchmark.

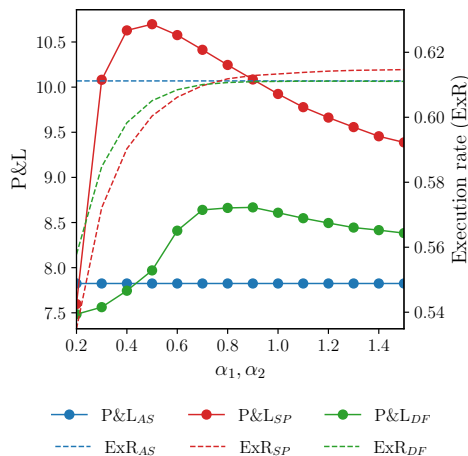


Figure 3: One-hour P&L and ExR as functions of  $\alpha_1$  and  $\alpha_2$

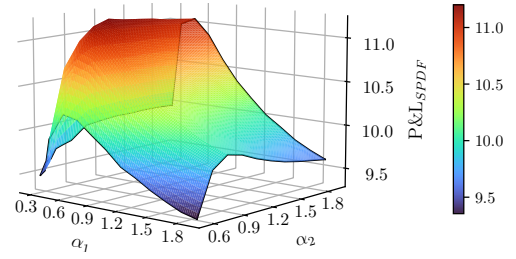


Figure 4: *SPDF* one-hour P&L as a function of  $\alpha_1$  and  $\alpha_2$

We find that  $P\&L_{SP}$  and  $P\&L_{DF}$  have their optima for  $\alpha_1 < 1$  and  $\alpha_2 < 1$ , respectively, indicating relatively high sensitivity to predictions. For very small  $\alpha_1$  and  $\alpha_2$ , *SP* and *DF* are little profitable as orders get filled rarely. For  $\alpha_1, \alpha_2 < 1$ , *SP* and *DF* outperform *AS* despite less order executions, suggesting larger spreads earned or reduced adverse selection cost. The optimal calibration for *SP* is  $\alpha_1 = 0.5$ , resulting in 35% P&L improvement over *AS*, and for *DF* it is  $\alpha_2 = 0.9$ , resulting in 10% P&L improvement.

Figure 4 illustrates the average one-hour P&L achieved by *SPDF* as a function of  $\alpha_1$  and  $\alpha_2$ . Similar to Figure 3, P&L is the lowest for large and very small values of  $\alpha_1$  and  $\alpha_2$ , and has its maximum for the combination of  $\alpha_1 = 0.55$  and  $\alpha_2 = 1.00$ , improving P&L by more than 40%. This calibration is similar to the stand-alone strategies *SP* and *DF*, requiring us to be about twice as sensitive to predictions when adjusting order depth to trade flow than when adjusting the reservation price to adverse price movements.

## 6 Trading simulation with order predictions

### 6.1 Prediction quality

We analyze prediction quality benchmarked against a single-layer feedforward neural network (SFN), which directly maps the inputs to  $R^b$  and  $R^a$ . Figure 5 illustrates the frequency distribution of  $\hat{R}^b$  (dashed) and  $\hat{R}^a$  (solid) over all test periods. The gray lines focus on  $R^b$  and  $R^a$ . Regarding the distribution of  $R^b$  and  $R^a$ , RNN apparently captures the true case better than SFN by achieving a stretched and rather flat predictions distribution. Both models predict very large and small values poorly, with substantially less (RNN) or hardly any (SFN) predictions in the upper and lower 10%.

Table 1 reports average prediction errors across all test periods aggregated by prediction error sign ('Sign'). Sign group '-' ('+') includes all negative (positive) prediction errors. We calculate prediction errors by  $\hat{R}^b - R^b$  and  $\hat{R}^a - R^a$ , respectively. Throughout all groups, RNN shows smaller prediction errors than SFN. For instance, if overpredicting percentile ranks, SFN ranks buy trades on average 23 percentile ranks too high, whereas RNN ranks them on average 14 ranks too high, suggesting an improvement by 40%. The mean squared error of RNN is less than half of the SFN model.

Next, we investigate whether prediction quality depends on

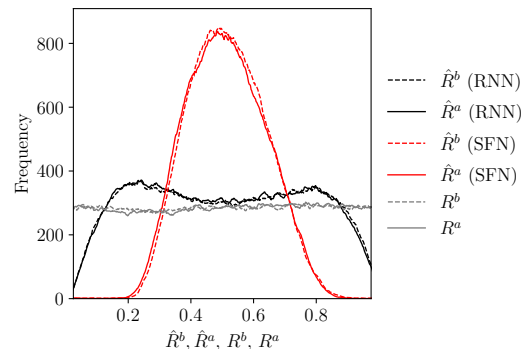


Figure 5: Frequency distribution of true and predicted values

| Side                     | $\hat{R}^b$ |      |       | $\hat{R}^a$ |      |       |
|--------------------------|-------------|------|-------|-------------|------|-------|
| Sign                     | -           | +    | All   | -           | +    | All   |
| Average prediction error |             |      |       |             |      |       |
| RNN                      | -0.14       | 0.14 | -0.00 | -0.14       | 0.14 | -0.00 |
| SFN                      | -0.22       | 0.23 | 0.01  | -0.22       | 0.22 | 0.00  |
| Mean squared error (%)   |             |      |       |             |      |       |
| RNN                      | 3.78        | 3.78 | 3.78  | 3.83        | 3.80 | 3.82  |
| SFN                      | 6.96        | 7.21 | 7.09  | 7.14        | 7.04 | 7.09  |

**Table 1:** Prediction error metrics

market sentiment. Analogous to the sequence length, we group the last ten seconds before a prediction into two equal-sized intervals, where the first (second) interval  $TFI_{1-5}$  ( $TFI_{6-10}$ ) represents the aggregated  $TFI$  of the first (last) five seconds. Figure 6 shows the mean squared error as a function of  $TFI_{1-5}$  and  $TFI_{6-10}$  for RNN and SFN predictions. Prediction models perform well when the preceding flow of buy and sell market orders is relatively balanced, and poorly when there is an excess of buyer- or seller-initiated trades. This behavior is less pronounced for RNN model than for SFN predictions, indicating that RNN outperforms SFN when making predictions based on imbalanced trade flows.

## 6.2 Trading performance

We conduct trading simulations separately for  $SP$ ,  $DF$  and  $SPDF$ , once based on RNN and once based on SFN predictions. Table 2 presents the equal-weighted average of cumulative P&Ls achieved within non-overlapping periods of 1 minute (min) to 12 hours (h). In the majority of periods, prediction-based strategies lead to an improvement over  $AS$ , e.g., we improve one-hour P&L by up to 7%. We also perform t-tests between the P&L distributions of prediction-based strategies and that of  $AS$  strategy and find that increases in P&L are statistically significant in most cases. RNN-based strategies produce, on average, a higher P&L than SFN-based strategies. Moreover, the standard deviation of P&Ls illustrate that higher profit chances come at the cost of higher risk. Only  $DF$  actually achieves a meaningful increase in profits while keeping standard deviation constant.

In order to quantify the P&L improvement, we define the P&L surplus as the difference between the P&L of a prediction-based strategy and the P&L of  $AS$  (P&L<sub>AS</sub>). The left chart in Figure 7 il-

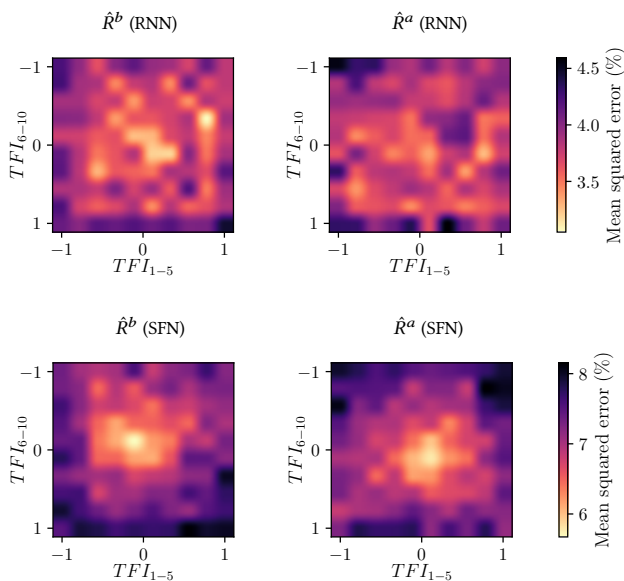
| Strategy |     | 1 min             | 5 min             | 1 h               | 12 h                 |
|----------|-----|-------------------|-------------------|-------------------|----------------------|
| $AS$     |     | 0.13<br>(0.25)    | 0.66<br>(0.79)    | 7.89<br>(5.71)    | 81.37<br>(32.95)     |
| $SP$     | RNN | 0.14***<br>(0.28) | 0.70***<br>(0.95) | 8.06**<br>(7.48)  | 101.93***<br>(44.71) |
|          | SFN | 0.14***<br>(0.27) | 0.68*<br>(0.92)   | 8.01*<br>(7.29)   | 91.86**<br>(42.27)   |
| $DF$     | RNN | 0.13**<br>(0.25)  | 0.69**<br>(0.80)  | 8.34***<br>(5.74) | 88.10**<br>(32.56)   |
|          | SFN | 0.13<br>(0.25)    | 0.66<br>(0.80)    | 7.88<br>(5.76)    | 81.00<br>(32.91)     |
| $SPDF$   | RNN | 0.14***<br>(0.28) | 0.75***<br>(0.93) | 8.45***<br>(7.29) | 104.13***<br>(43.61) |
|          | SFN | 0.14**<br>(0.26)  | 0.67<br>(0.90)    | 7.92<br>(7.08)    | 92.01**<br>(41.06)   |

Note: Numbers in brackets report the P&Ls' standard deviation; \*\*\*, \*\* and \* indicate significant difference from  $AS$  at the 1%, 5% and 10% level.

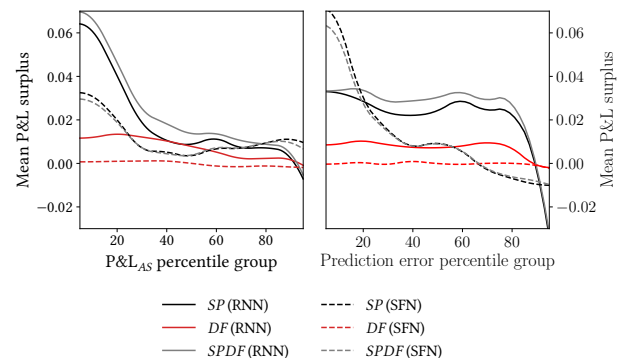
**Table 2:** Cumulative P&L by trading duration

lustrates the average one-minute P&L surplus as a function of one-minute P&L<sub>AS</sub> percentile groups, where percentile group 1 (100) contains the worst (best) 1% of all one-minute P&L<sub>AS</sub>. In both RNN- and SFN-based strategies, we benefit from prediction-based strategies when  $AS$  performs poorly. The better  $AS$  performs, the less beneficial or more detrimental prediction-based strategies are. The right chart plots the P&L surplus over one-minute prediction error percentile groups, where percentile group 1 (100) contains the 1% smallest (largest) one-minute sums of absolute prediction errors for both buy and sell forecasts. The graph illustrates that P&L surplus is large (small) when prediction errors are small (large), suggesting that prediction quality is an indicator of P&L performance. Besides, the charts show that  $DF$  is less dependent on prediction quality and P&L<sub>AS</sub> than  $SP$  or  $SPDF$  strategy.

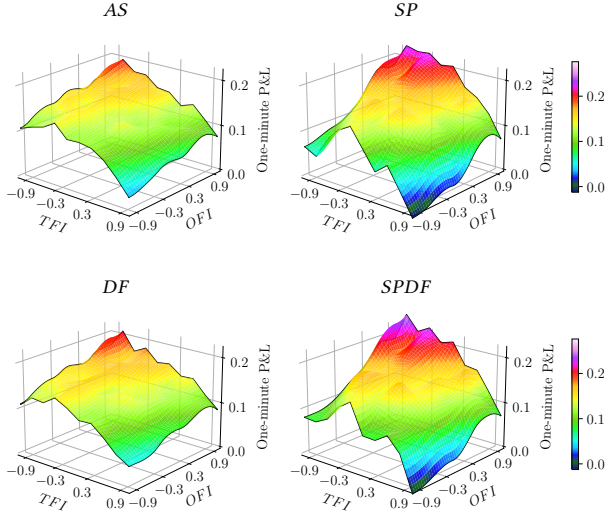
Figure 8 illustrates each strategy's average one-minute P&L as a function of one-minute  $TFI$  and one-minute  $OFI$ . We calculate one-minute  $TFI$  based on minute-by-minute aggregated buy and sell trades, whereas for one-minute  $OFI$ , we consider equal-weighted averages of all one-second LOB records within a minute. Trading is least profitable during a seller overhang in the LOB ( $OFI < -0.5$ ) in combination with a buyer surplus in trade flow ( $TFI > 0.5$ ), potentially representing high competition among waiting sellers such that we can catch only a small share of impatient buyers. P&L is the highest in the opposite scenario,  $OFI > 0.5$  with  $TFI < -0.5$ , indicating that we are more successful in competing with buyers than with sellers (see Section 6.3). Overall,  $AS$  and  $DF$  ( $SP$  and  $SPDF$ ) are less (more) dependent on market dynamics and therefore provide a less (more) volatile P&L profile.  $SP$  and  $SPDF$  are superior to  $AS$  and  $DF$  only if  $OFI > 0$



**Figure 6:** Mean squared prediction error by market sentiment



**Figure 7:** One-minute P&L surplus over percentile groups

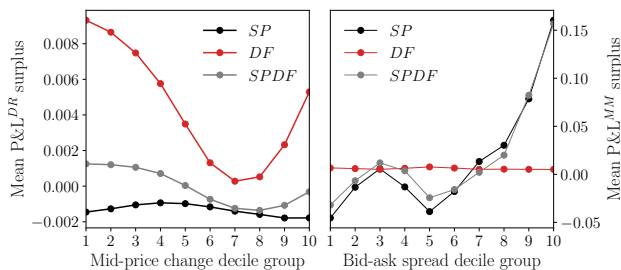


**Figure 8:** One-minute P&L as a function of  $TFI$  and  $OFI$

and  $TFI < 0$ , whereas in all other situations  $AS$  and  $DF$  are superior. Hence, dealers could switch between trading strategies in response to changes in trade and order flow. e.g., switch to  $AS$  or  $DF$  ( $SP$  or  $SPDF$ ) when the buyer share (seller share) in trade flow is large while the buyer share (seller share) in the LOB is small.

Since P&L includes both market making revenues and unrealized directional changes of the net position [60], we next aim to understand each component's contribution to P&L. We calculate P&L generated from directional movements,  $P\&L^{DR}$ , as the change in value of the net position caused by the mid-price change [4], i.e., if the mid-price increases by 0.1 while we are long 2 assets, it is  $P\&L^{DR} = 0.2$ . The left plot in Figure 9 reports RNN-based  $P\&L^{DR}$  surplus as a function of one-minute mid-price change decile groups, i.e., decile group 1 (10) contains the 10% minutes with the most negative (positive) mid-price changes. In case of  $DF$ ,  $P\&L^{DR}$  surplus is the highest (smallest) when the mid-price sharply (slightly) increases or decreases. As  $P\&L^{DR}$  can be viewed as an approximation of adverse selection cost, this result suggests that  $DF$ 's reduction in adverse selection cost is proportional to the magnitude of price movement: The larger the positive or negative price movement, the larger the reduction in adverse selection cost.

As second P&L component, we calculate P&L from market making activities,  $P\&L^{MM}$ , by subtracting  $P\&L^{DR}$  from P&L [60]. Thus, we implicitly subsume realized directional revenues [4] under market making revenues. The right plot in Figure 9 shows average  $P\&L^{MM}$  surplus as a function of average one-minute bid-ask spread decile groups. Both  $SP$  and  $SPDF$  achieve the largest increase in market making revenues when bid-ask spreads are large, suggesting that the strategies are most successful at adjusting their LOB position to trade flow when the order book is less liquid. From



**Figure 9:** One-minute  $P\&L^{DR}$  surplus and  $P\&L^{MM}$  surplus

both charts in Figure 9, we infer that P&L surplus of  $DF$  is mainly driven by reduced adverse selection cost, while P&L surplus of  $SP$  and  $SPDF$  is mainly due to increased market making revenues.

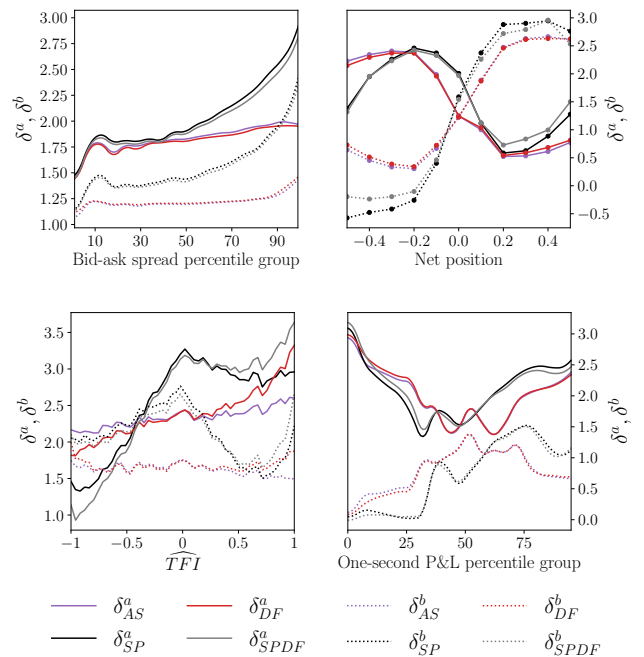
### 6.3 Limit order book dynamics

Finally, we analyze how our strategies affect our position in the LOB. Figure 10 illustrates the average  $\delta^a$  (solid lines) and the average  $\delta^b$  (dotted lines) as functions of bid-ask spread percentile groups (top left), the net position (top right), the predicted trade flow imbalance (below left), and one-second P&L percentile groups (below right).

The top left graph shows that  $SP$  and  $SPDF$  place orders deeper in the book than  $AS$  and  $DF$ , suggesting that  $SP$  and  $SPDF$  seek increasing the margin through wider spreads rather than increasing trade volume through tighter spreads.  $AS$  and  $DF$  quotes are less sensitive to bid-ask spreads, i.e., when bid-ask spreads increase,  $AS$  and  $DF$  follow this trend slower than  $SP$  and  $SPDF$ . It is  $\delta^a > \delta^b$  in most cases, indicating that our average net position is short (see Figure 15 in the Appendix). Thus, despite prediction-based corrections, the dominant parameter determining  $\delta^a$  and  $\delta^b$  remains  $q$ . Correspondingly, the top right graph shows that  $\delta^a$  is large (small) when we are short (long) and  $\delta^b$  is small (large) when we are short (long) to return to a neutral net position.

The chart below left illustrates  $DF$ 's and  $SPDF$ 's protection against adverse price selection:  $\delta^a$  increases for  $\overline{TFI} > 0$  to protect against informed buyers, and decreases for  $\overline{TFI} < 0$  to incentivize short exposure for positively participating in the price decline. Accordingly, buy prices are lower represented by increasing  $\delta^b$  when predicting a sell surplus, although the effect is less pronounced than for sell orders, which could be due to the fact that we are short on average and thus generally place buy orders deeper in the book. The chart below right illustrates that submitting sell orders deep in the book is risky: The highest and lowest P&L are associated with the deepest sell orders, while sell orders close to the top of the order book are associated with average P&L. Again, this relationship is less clear for buy orders.

Next, we compare the price level priority of orders between prediction-based strategies and the  $AS$  strategy. We start counting



Note:  $SP$ ,  $DF$  and  $SPDF$  use RNN predictions.  $\delta^a, \delta^b$  denote in basis points.

**Figure 10:** Depth of orders in the order book

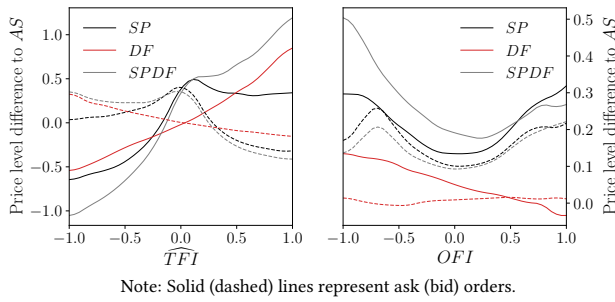


Figure 11: Price level difference to AS

price levels at the top of the book as position 1, continuing with the second-best price level as position 2, and so on. Thus, a position difference of 1 (−1) means that the prediction-based strategy is one price level behind (ahead of) AS, i.e., it has a one price level lower (higher) priority. Figure 11 shows the average price level difference for RNN-based strategies as a function of  $TFI$  (left) and  $OFI$  (right). The left chart shows that all three strategies post sell (buy) orders deeper in the book when predicting a price increase (decline) to protect against informed buyers (sellers). This effect is most pronounced for  $DF$  and  $SPDF$ , since  $SP$  initially does not consider directional forecasts. The right chart illustrates that  $SP$  and  $SPDF$  orders have a lower (equal) priority when there is either (neither) a large buy or (nor) a large sell overhang in the order book. In contrast,  $DF$  posts sell orders closer to the top of (deeper in) the book when  $OFI > 0$  ( $OFI < 0$ ), thus acts contrary to  $TFI$ .

To understand how our positioning in the LOB contributes to P&L, the bar chart in Figure 12 illustrates the execution rate (left y-axis) by price level. An execution rate of 30% at price level 3 means that 30% of our orders queuing at price level 3 are filled within five seconds. The circle markers indicate the average margin earned (right y-axis), measured by  $\delta^a + \delta^b$ . A margin of 5 basis points at price level 2 means that we earn 5 basis points on average if one buy and one sell order, each waiting at the second best price level, get simultaneously filled by market orders.

While at the top of the order book all strategies achieve similar execution rates and margins,  $SP$  and  $SPDF$  exceed  $AS$  and  $DF$  for higher price levels. The margin advantage suggests that  $SP$  and  $SPDF$  can execute deeper orders even when the bid-ask spread is large, whereas  $AS$  and  $DF$  execute on these levels only when the bid-ask spread is small anyway and, accordingly, earn less margin. Thus,  $SP$  and  $SPDF$  maximize the margin earned by increasing the depth of orders while keeping the execution share at a high level. Figure 13 shows each strategy's share of executed orders by price level, illustrating that  $SP$  and  $SPDF$  execute more orders at price levels 2 to 5 than  $AS$  and  $DF$ . We infer that the higher market making revenue of  $SP$  and  $SPDF$  largely comes from a higher execution share of orders with more profitable spreads.

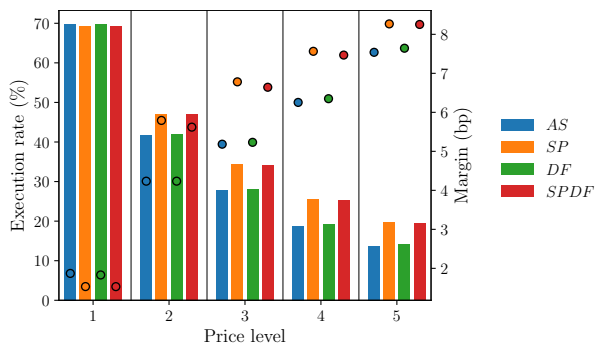


Figure 12: Trade executions by price level

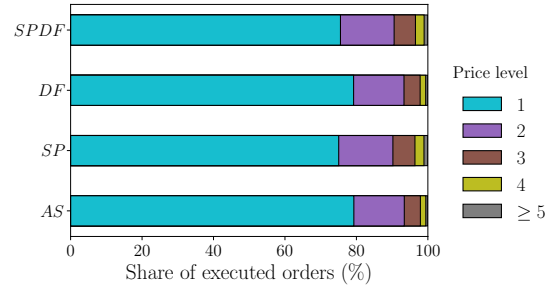


Figure 13: Share of order executions by price level

## 7 Conclusion

In this paper, we present a data-driven attempt to overcome dealers' trade-off between volume and margin. To this end, we develop a sequence prediction model for buyer- and seller-initiated trades that predicts how the trade sizes of the next five seconds relate to the trade sizes of the previous 24 hours. Using the Avellaneda-Stoikov model as an example, we show how to adjust reservation price and order depth in response to predictions based on empirically calibrated sensitivity parameters. In a trading simulation, we find both higher market making revenues through trade flow-optimized positioning in the order book and lower adverse selection costs through anticipated price movements.

For future work, we see two possible approaches to improving our method. Given the recent development of reinforcement learning in high-frequency trading, one approach would be to incorporate our predictions into the training of an agent, thus providing predictive information to the agent. Second, instead of approximating price changes using trade flow imbalance, predicting price changes directly could improve our protection against adverse selection as negative (positive) trade flow imbalance need not always be associated with a price decrease (increase).

## 8 Appendix

Figure 14 presents the average number of price levels consumed by  $R^b$  and  $R^a$ . The minimum price level consumed is 1, i.e., the order has been fully matched with open limit orders at the top price level. The figure indicates that larger trades are associated with larger spreads earned by dealers on average.

Figure 15 shows the percentiles of net positions held as an equal-weighted average for both ETH-USD and BTC-USD pairs. The inventory limit is  $\pm 0.5$  BTC or  $\pm 0.5$  ETH, respectively. In more than 50% (less than 20%) of all seconds, the dealer is net short (net long), whereas in about 30% of all seconds the position is neutral.

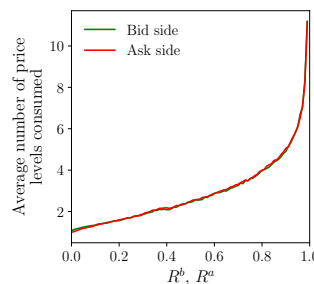


Figure 14: Price level consumption by  $R^b$  and  $R^a$

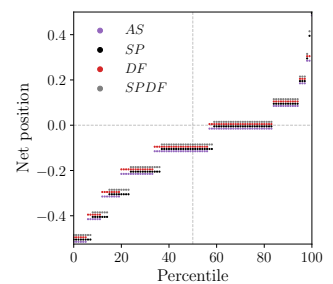


Figure 15: Percentiles of net positions

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